How Adaptive Traffic Models Redistribute Fragility: A Multi-Shift Diagnostic of Spatiotemporal Architectures

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Abstract

Spatiotemporal graph neural networks (STGNNs) dominate traffic forecasting benchmarks, yet which architectural properties govern their robustness under realistic deployment shifts remains unclear. Eight frozen architectures are evaluated under three deployment-motivated shifts—inference-time sensor dropout, zero-shot cross-city transfer (METR-LA $\rightarrow$ PEMS-BAY), and weekday/weekend temporal subgroup shift—with all predictions pre-registered before any experiments ran.

Robustness turns out to be shift-specific, and architectural progress changes failure modes rather than eliminating them. Adjacency routing concentration, $\mathrm{max_mass}(A)$, explains the sensor-dropout ranking at Pearson $\rho{=}{+}0.979$ across all four adaptive models. Graph WaveNet's dense softmax routing ($\mathrm{max_mass}{\approx}0.23$) degrades $4.3\times$ more than DCRNN under 30% dropout; a mask-aware inference variant that zeroes dropped-sensor adjacency entries recovers 75% of the gap without retraining. Sensor-dropout robustness is therefore not a property of topology learning per se, but of routing density: AGCRN ($\mathrm{max_mass}{=}0.008$) and STAEformer show fragility comparable to DCRNN despite learning adaptive spatial relations. Cross-city and temporal fragility lie on orthogonal axes: MegaCRN achieves the best zero-shot city-transfer score yet exhibits 8–9$\times$ larger temporal subgroup gaps than the canonical bank—a fragility axis invisible in five-model evaluations and not driven by explicit temporal covariates in MegaCRN (mechanism ablations, §5). Deployment-oriented model selection should be based on the target shift type and routing density of candidate architectures, not on in-distribution accuracy alone.

1 Introduction

Traffic forecasting underpins urban services—ride-hailing dispatch, signal timing, congestion management—and STGNNs are the dominant tool [Li et al., 2018, Wu et al., 2019, Yu et al., 2018]. In the lab, these models are evaluated on held-out data from the same city, sensors, and time window as training. Deployment conditions differ: sensors fail, models trained in one city are applied to another, and traffic shifts between weekdays and weekends. Which architectures hold up under these shifts is a deployment question that in-distribution accuracy cannot answer.

Prior STGNN robustness work emphasizes synthetic corruptions [Zhang et al., 2025] or frames cross-city transfer as few-shot learning [Zeng et al., 2025, Jing et al., 2026]. Missing is a controlled *diagnostic* study that (i) uses deployment-realistic shifts, (ii) evaluates the same frozen backbone under multiple shifts, (iii) attributes differences to specific architectural choices, and (iv) pre-registers hypotheses so refuted predictions carry the same evidential weight as confirmed ones. Cross-city comparisons also conflate transfer difficulty with inductive bias—adaptive models can re-learn target

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topology, fixed models cannot—which a matched-structural-access protocol (source-city topology for all backbones) removes. No prior study systematically compares how different adaptive-relation designs—GWNet's dense softmax routing, MegaCRN's meta-graph memory, STAEformer's adaptive embeddings—redistribute fragility across deployment shifts.

We assemble a bank of eight architectures trained on METR-LA—a five-model canonical bank spanning graph-free (GRU), fixed-graph (DCRNN, STGCN), and adaptive-graph (MTGNNLight, GWNet), plus three modern adaptive-relation models (AGCRN, MegaCRN, STAEformer)—and evaluate each frozen checkpoint under three shifts: **sensor dropout** (10–50% zeroed at inference, 5 random masks), **cross-city transfer** (zero-shot METR-LA $\rightarrow$ PEMS-BAY under matched structural access), and **temporal shift** (weekend vs. weekday windows within the METR-LA test set).

Three findings stand out:

1. No architecture dominates all three shifts; rankings change qualitatively between shift types. GWNet's dense adaptive adjacency is most fragile under sensor dropout ($\sim 4.3\times$ worse than DCRNN at 30%) but not the worst at city transfer. DCRNN leads cross-city in the canonical bank; MegaCRN overtakes it once the extended set is added.

2. GWNet's dropout fragility is the visible tail of a routing-density relation: $\mathrm{max_mass}(A)$ ranks Δ_{sensor} at Pearson $\rho{=}+0.979$ over four adaptive backbones on METR-LA, replicating at $\rho{=}+0.910$ on PEMS04 with native checkpoints. Replacing the learned adjacency with a fixed road graph cuts dropout sensitivity by 46%; a mask-aware variant recovers 75% without retraining. AGCRN ($\mathrm{max_mass}{=}0.008$) and STAEformer match DCRNN despite learning adaptive relations: the operative factor is routing density, not topology learning.

3. Modern adaptive-relation models distribute their failures across axes. AGCRN and STAEformer survive sensor dropout; MegaCRN leads cross-city. But MegaCRN and STAEformer carry weekday/weekend gaps $\sim$8–9$\times$ the canonical bank—an axis that the five-model evaluation could not have surfaced, and one that explicit TOD/DOW covariates do not explain (Section 5).

Sixteen predictions were pre-registered in five stages, each frozen before the next round of training (ledger and confirmed/refuted/partial criteria: Appendix A). Four were refuted: GWNet's dominant failure axis was predicted to be cross-city transfer, not sensor dropout; temporal fragility was predicted to be architecture-invariant, but is not—findings invisible in a confirmation-only evaluation. Three follow-ups—PEMS04-native routing-density replication (Appendix O), the mask-aware dropout-ratio sweep (Appendix P), and cross-city zero-padding sensitivity (Appendix Q)—are flagged as reviewer-driven, reported alongside the pre-registered core. The frozen-checkpoint protocol is reproducible end-to-end on a single commodity GPU; checkpoints are released as supplementary material.

2 Related Work

Spatiotemporal graph neural networks for traffic. DCRNN [Li et al., 2018] introduced bidirectional diffusion convolution inside a recurrent encoder-decoder. Graph WaveNet [Wu et al., 2019] replaced the fixed graph with a learned adaptive adjacency. STGCN [Yu et al., 2018] alternates gated temporal convolutions with spectral graph convolutions. A plain GRU is the graph-free baseline; MTGNNLight adds a sparse adaptive topology. These five form our canonical bank. Inductive alternatives like MTGNN [Wu et al., 2020] and AGCRN [Bai et al., 2020] infer graph structure from data, while newer adaptive-relation models—MegaCRN [Jiang et al., 2023] (meta-graph memory with temporal covariates) and STAEformer [Liu et al., 2023a] (Transformer with spatial-temporal adaptive embedding)—now top traffic leaderboards. We include the MTGNN-inspired light variant in the canonical evaluation, and extend the diagnostic to AGCRN, MegaCRN, and STAEformer for external validity. Our results show that MTGNNLight's sparser learned graph reduces but does not eliminate sensor-dropout fragility relative to GWNet's dense softmax product graph, while AGCRN and STAEformer match DCRNN under sensor dropout—which points to adjacency density and routing design (not topology-learning per se) as the source of GWNet's routing brittleness. None of these architectures were previously evaluated for robustness under deployment-time distribution shift; the published numbers are in-distribution held-out accuracy only.

Distribution shift in spatiotemporal models. A handful of recent methods address distribution shift in STGNNs. STRAP [Zhang et al., 2025] handles spatio-temporal OOD scenarios with retrieval-augmented pattern learning: at inference, it retrieves historical patterns from a stored library and injects them via a plug-and-play prompt, adapting the model without re-training. TL-GPSTGN [Jing et al., 2026] prunes graph context with information-theoretic criteria during training to improve OOD generalization. These propose interventions; our work provides a diagnostic—which backbones degrade under which shifts and why—a prerequisite for designing targeted retrieval, topology-aware, or imputation-based responses.

Cross-city transfer learning. Several papers frame cross-city generalization as few-shot or federated learning. FedTT [Zeng et al., 2025] trains a federated model across multiple cities with traffic-view imputation and a domain adapter for cross-city discrepancies. CAST-CKT [Fofanah et al., 2026] uses chaos-aware attention and adaptive topology learning for cross-city knowledge transfer. The LargeST benchmark [Liu et al., 2023b] ships with a traffic-pattern bank retrieved by pattern matching. All of these use some target-city data at training or adaptation time. We instead evaluate the zero-shot case: models trained only on METR-LA applied directly to PEMS-BAY, no adaptation—the setting that arises whenever a target city has no historical data.

Topology-agnostic and foundation-model approaches. UniSTOK [Anonymous, 2025] tokenizes spatiotemporal signals uniformly across variable sensor configurations; topology-free foundation models like Chronos-2 [Ansari et al., 2024] discard the graph entirely. Cross-city foundation models train on many cities and adapt by fine-tuning, which again needs target-domain data. Our zero-shot setting is complementary: it asks which architectural properties confer robustness before any adaptation is available—a prerequisite for deciding when topology-agnostic or fine-tuning approaches are worth the trouble.

Sensor failure and data imputation. Mask-aware reconstruction architectures like GRIN [Cini et al., 2022] and BRITS [Cao et al., 2018] explicitly reconstruct missing inputs before or during graph aggregation. We ask the orthogonal question: among standard STGNNs without imputation, which architectural choices confer robustness and which amplify fragility? The answer reveals architecture-intrinsic failure modes that imputation-based and foundation-model approaches would otherwise mask. Our LOCF baseline (Appendix) confirms that temporal carry-forward largely eliminates the fragility ranking, so the sensor-dropout findings are specific to the silent zero-masking scenario.

3 Problem Setup and Backbone Bank

3.1 Backbone Architectures

The bank covers five architectures from graph-free to fixed-graph to adaptive-graph.

DCRNN [Li et al., 2018] treats traffic as a diffusion process on a fixed road-distance adjacency $\mathbf{A}$, with bidirectional updates $\mathbf{h}_{t+1} = f\left(\sum_{k=0}^{K}(\mathbf{A}^k \oplus (\mathbf{A}^\top)^k)\mathbf{x}_t,\ \mathbf{h}_t\right)$; the adjacency is fixed at training and inference. **Graph WaveNet (GWNet)** [Wu et al., 2019] adds a learned adaptive adjacency

$$\hat{\mathbf{A}} = \text{softmax}(\mathbf{E}_1\mathbf{E}_2^\top), \tag{1}$$

with $\mathbf{E}_1, \mathbf{E}_2 \in \mathbb{R}^{N\times d_e}$ trained end-to-end and tied to the N training nodes. **STGCN** [Yu et al., 2018] alternates gated temporal convolutions with truncated-Chebyshev spectral graph convolutions over the same fixed road adjacency. **GRU** is a graph-free recurrent baseline that treats the N sensors as a single feature vector and uses no adjacency. **MTGNNLight** is a lightweight MTGNN variant [Wu et al., 2020] that learns a sparse adaptive graph via pairwise cosine similarity on a single embedding $\mathbf{E} \in \mathbb{R}^{N\times d}$ and keeps only top-k neighbours per node—a direct contrast to GWNet's dense all-pairs softmax that isolates whether dropout fragility depends on adjacency density or on topology-learning per se (P14).

All five share a unified config (hidden dim 64, seq/pred 12, Adam 1×10^{-3}, weight decay 10^{-4}, cosine schedule, 100 epochs, patience 15). To rule out training-artifact explanations of GWNet's fragility we additionally train **GWNet[CM]** (patience 30) and **GWNet[Strong]** (P17; hidden 128, adaptive embed 20, patience 30); both are frozen at shift-eval time.

3.2 Backbone Bank Protocol

We train each backbone on METR-LA [Liu et al., 2023b] (207 sensors, 34,272 timesteps at 5-min, $\sim$4 months, chronological 70/10/20 split) with three seeds (0,1,2). Checkpoints are frozen after training; no re-training occurs at shift-eval time. In-distribution MAE (mph, denormalized): DCRNN 6.57 ± 0.03, MTGNNLight 6.88 ± 0.11, STGCN 7.12 ± 0.05, GWNet[Strong] 7.42 ± 0.04, GRU 7.49 ± 0.12, GWNet[CM] 7.83 ± 0.10, GWNet 7.85 ± 0.11. GWNet underperforms DCRNN and even MTGNNLight on in-distribution data despite its adaptive adjacency—a limitation discussed in §6.

3.3 Extended adaptive-relation models

To probe generality, we add three architectures (same METR-LA data, same split, 3 seeds, frozen after training): **AGCRN** [Bai et al., 2020] (node-adaptive GRU, no temporal covariates); **MegaCRN** [Jiang et al., 2023] (meta-graph memory with TOD/DOW inputs); **STAEformer** [Liu et al., 2023a] (Transformer with spatial-temporal adaptive embeddings encoding temporal covariates). In-distribution masked MAE (mph, excluding zero-speed targets): AGCRN 4.09 ± 0.02, MegaCRN 3.04 ± 0.06, STAEformer 2.93 ± 0.01. These use masked MAE, the canonical bank uses plain MAE; we avoid direct cross-family clean-performance ranking.

3.4 Distribution shifts

Sensor dropout. At inference, a random 30% of sensor channels are zero-masked. The 207-node model is not re-instantiated; it sees a 207-dimensional input with zeroed channels, and MAE is computed on the kept sensors only so clean and masked evaluations are directly comparable. Five independent random masks per checkpoint separate mask-sampling from model-weight variance; Tables 1–3 report the primary-mask aggregate (all 5-mask aggregates are within ± 0.005). Structured dropout patterns (hub and contiguous) are in Appendix B. This models silent sensor failure.

Cross-city transfer. The frozen METR-LA model is evaluated zero-shot on PEMS-BAY [Liu et al., 2023b] (Bay Area, 325 sensors, 52,116 timesteps). A second pair (PEMS04 source, 307 sensors; zero-shot on METR-LA and PEMS-BAY) is in Appendix K for cross-city generality (P18). "Zero-shot" means no re-training or fine-tuning. Because PEMS-BAY has 325 sensors and our models capacity 207, we subsample 207 PEMS-BAY sensors (5 subsets); rankings are stable at 150-node coverage (Appendix C).

Matched structural access (canonical comparison). The main cross-city comparison uses source-only adjacency: every backbone uses the METR-LA adjacency at inference, removing the asymmetry that adaptive-graph models cannot access target topology by construction. Under this constraint, any observed advantage is architectural, not an artefact of privileged topology access (target-adj results: Appendix H). MAE is normalized with source-city statistics for cross-architecture comparability.

Temporal shift. METR-LA spans weekdays and weekends with different traffic patterns; the model is trained on both. We stratify the test set by day type—weekday (5,104 windows) and weekend (1,728 windows)—and report $\Delta_{\text{temp}} = \text{MAE}_{\text{weekend}} - \text{MAE}_{\text{weekday}}$. This is a subgroup performance gap within the existing test split, not deployment to a new time window. It requires no additional data and reveals whether graph structure helps one traffic regime more than the other.

Protocol summary. Train on METR-LA (207 sensors, 5-min, 70/10/20 split); 3 seeds, 100 epochs, patience 15; frozen at evaluation. *Sensor:* 30% silent zero-mask, 5 masks, MAE on kept sensors. *City:* zero-shot to PEMS-BAY (325→207 subsampled, 5 subsets), source-only adjacency (matched-structural-access; padding sensitivity in Appendix Q). *Temporal:* weekday (5,104) vs. weekend (1,728) test windows. *Aggregation:* mean $\pm$ sample-std over 3 seeds $\times$ 5 masks; METR-LA train-statistics normalization for the canonical bank, model-specific scaler for extended models (§3.3).

4 Robustness Taxonomy

Pre-registered hypotheses. We pre-registered 16 predictions in five stages before running the corresponding experiments. Of these, 10 were confirmed, 4 were refuted, and 2 were partially

Table 1: **Robustness rankings are shift-specific: no single backbone dominates.** Five backbones evaluated across three natural shifts (3 seeds, mean ± std). **Sensor dropout** Δ_{sensor}: MAE increase on kept sensors after 30% zero-masking (5 masks). **Cross-city**: normalized MAE on PEMS-BAY under matched structural access (source-only adjacency for all models; GWNet/MTGNNLight are source-tied by construction). Target-adj mph values in Appendix H. **Temporal** Δ_{temp}: weekend minus weekday MAE (normalized). Lower is better for all columns; shift deltas in normalized units (METR-LA training statistics). Backbones ordered by sensor-dropout robustness.

Backbone	In-dist MAE (mph)	Δ_{sensor} (30%, norm.)	X-city (norm.)†	Δ_{temp} (norm.)
DCRNN (fixed diffusion)	6.57 ± 0.03	$+0.021 \pm 0.002$	0.180 ± 0.001	$+0.087 \pm 0.004$
MTGNNLight (sparse cosine-sim)	6.88 ± 0.11	$+0.051 \pm 0.012$	0.244 ± 0.005	$+0.075 \pm 0.007$
STGCN (fixed spectral)	7.12 ± 0.05	$+0.078 \pm 0.007$	0.249 ± 0.010	$+0.094 \pm 0.003$
GRU (graph-free)	7.49 ± 0.12	$+0.025 \pm 0.012$	0.278 ± 0.011	$+0.093 \pm 0.003$
GWNet (dense adaptive adj)	7.85 ± 0.11	$+0.088 \pm 0.017$	0.268 ± 0.007	$+0.094 \pm 0.009$

†Source-only adjacency; METR-LA-stat normalized MAE on PEMS-BAY.

confirmed. The full prediction–outcome ledger is in Appendix A; key confirmed predictions (P14–P19, P21) are referenced inline where their results appear.

Table 1 reports the main results across all five backbones and three shift types.

Sensor dropout. Best to worst (5 masks, 3 seeds): DCRNN $+0.021$, GRU $+0.025$, MTGNNLight $+0.051$, STGCN $+0.078$, GWNet $+0.088$. GWNet degrades $4.2\times$ DCRNN's gap ($>7\times$ cross-seed std). MTGNNLight's sparse cosine-similarity adjacency (top-k=10) is 42% less fragile than GWNet's dense softmax, which points at adjacency density rather than topology-learning per se. A capacity-scaled **GWNet[Strong]** halves the gap to $+0.050$ but is still $2.5\times$ DCRNN; the full 12-config sweep GWNet[Best] sits at $2.1\times$ (Table 2; Appendix E.2 for per-mask values).

Cross-city transfer under matched structural access. Under source-only adjacency, **DCRNN leads the canonical bank** (0.180 ± 0.001, a 49% gap over GWNet's 0.268). Fixed adjacency does not automatically buy transfer robustness: STGCN (0.249) nearly matches GWNet under fair comparison; its earlier target-adj advantage (0.202) was a topology-access artefact. DCRNN's bidirectional diffusion generalizes across topologies even when confined to source structure, and the advantage holds on a second city pair (Table 2). Index-repetition padding (used when target $N <$ source N) flatters graph-free and dense-adaptive backbones by $\sim$0.7 mph and fixed-graph backbones by $\sim$0.15 mph, but the ranking is unchanged and DCRNN remains best in both padded and no-pad evaluations (Appendix Q).

Temporal subgroup shift (canonical bank). Within the canonical bank, every model shows a weekend/weekday gap of $+0.087$ to $+0.094$ normalized MAE, with inter-architecture differences below one cross-seed standard deviation. None of these architectures condition on explicit time-of-day or day-of-week, so graph structure neither helps nor hurts on this within-dataset subgroup shift. §4.1 shows this near-invariance breaks once temporal-covariate-conditioned models are added.

Shift-specificity (canonical bank). Rankings differ across shift types: in-distribution accuracy does not predict sensor-dropout fragility, and fixed adjacency does not guarantee transfer robustness. DCRNN leads city transfer within the canonical bank (0.180), but §4.1 shows that fact does not survive the addition of modern adaptive-relation models. The central observation is shift-specificity: no single "most robust" backbone, and the right choice depends on the anticipated shift. Table 2 consolidates supplementary results supporting these claims.

4.1 External validity: modern adaptive-relation models

The canonical bank covers four spatial-aggregation strategies but omits the newer adaptive-relation designs that dominate traffic forecasting leaderboards. To test whether the findings carry over, we add three architectures (AGCRN, MegaCRN, STAEformer) trained on the same METR-LA data with the same split, frozen after training, evaluated under the same protocol. Table 3 reports results alongside DCRNN as a reference.

Table 2: **Key supplementary results confirming main findings.** Each row summarizes an appendix experiment that directly supports a main-text claim.

Experiment	Key result	Ref.
GWNet[Best] 12-config sweep (P21)	$\Delta_{\text{sensor}} = +0.042$ (2.1× DCRNN); optimal training cannot eliminate routing fragility	App. J
PEMS04 cross-city (P18)	DCRNN best in both directions: 7.01/2.65 mph vs next-best 11.03/4.58 mph	App. K
Source-adj vs target-adj	STGCN advantage disappears under fair comparison (0.249 src-only vs 0.202 target-adj)	App. H
Hub dropout (structured)	GWNet $+45\%$ worse ($+0.127$ vs. matched $+0.088$); DCRNN -62% better ($+0.008$)	App. B
Per-horizon city MAE (P19)	GWNet: $+4\%$ h1→h12 growth vs DCRNN $+46\%$—static mis-routing, not accumulation	App. L

PEMS04 cross-city values in mph (denormalized MAE); all Δ values in normalized units (METR-LA training statistics).

Table 3: **Modern adaptive-relation models exhibit heterogeneous failure profiles that do not support a family-wide fragility claim.** Clean performance uses masked MAE (excluding zero-speed targets), the standard convention for these architectures; DCRNN uses plain MAE and is included for shift-delta comparison only. Δ_{sensor}: normalized MAE increase at 30% zero-mask dropout (5 masks, 3 seeds). Cross-city: source-stat normalized MAE on PEMS-BAY (same protocol as Table 1). Δ_{temp}: weekend minus weekday normalized MAE (source-stat normalized units, same convention as Table 1).

Model	Clean MAE (mph)	Δ_{sensor} (30%, norm.)	Cross-city (src, norm.)	Δ_{temp} (norm.)
AGCRN	$4.09 \pm 0.02^{\dagger}$	$+0.019 \pm 0.001$	0.206 ± 0.011	$+0.061 \pm 0.008$
MegaCRN	$3.04 \pm 0.06^{\dagger}$	-0.048 ± 0.027	0.163 ± 0.004	$+0.734 \pm 0.019$
STAEformer	$2.93 \pm 0.01^{\dagger}$	$+0.016 \pm 0.001$	0.223 ± 0.009	$+0.750 \pm 0.003$
DCRNN (ref)	6.57 ± 0.03	$+0.020 \pm 0.002$	0.180 ± 0.001	$+0.087 \pm 0.004$

† Masked MAE (target > 0); DCRNN uses plain MAE. Cross-family clean-performance ranking is not meaningful due to this convention difference.

Sensor dropout. AGCRN ($+0.019$) and STAEformer ($+0.016$) match DCRNN ($+0.020$), so GWNet's fragility ($+0.088$) does not generalize to every adaptive-graph model. The distinction is dense all-pairs softmax (Eq. 1) vs. node-specific or attention-mediated relations. MegaCRN's negative delta (-0.048) is non-monotonic in dropout ratio and is adjacency-mediated smoothing toward a weekday prior (§5.3), not regularization.

Cross-city transfer. MegaCRN attains the best source-stat cross-city MAE (0.163), overturning the canonical-bank result that DCRNN (0.180) is the strongest zero-shot transfer backbone; AGCRN (0.206) and STAEformer (0.223) fall between DCRNN and GWNet.

Temporal subgroup shift. MegaCRN ($+0.734$) and STAEformer ($+0.750$)—both conditioning on time-of-day and day-of-week—exhibit normalized weekday→weekend gaps 8–9× DCRNN's ($+0.087$), consistent across seeds (std <0.02).[1] AGCRN ($+0.061$) does not condition on temporal covariates and matches the canonical bank. Yet direct covariate randomization or swap does not eliminate MegaCRN's gap (§5.3): the fragility lives in the memory bank or encoder dynamics, not the covariate channel.

Synthesis: fragility redistribution. Modern adaptive-relation models distribute their failures heterogeneously (Figure 1): AGCRN and STAEformer match DCRNN under sensor dropout, MegaCRN leads cross-city, and MegaCRN/STAEformer introduce a new temporal-subgroup axis. Architectural

[1]The 8–9× multiplier is in normalized units; in absolute masked mph, MegaCRN's and STAEformer's clean MAE ($\approx$ 3 mph) is roughly half DCRNN's (6.57 mph), so per-node-std normalization amplifies the visible ratio. Per-seed weekday/weekend mph values are in Appendix R (Table 26 caption).

Table 4: **Replacing GWNet's adaptive adjacency with a fixed road graph halves sensor-dropout sensitivity and improves city transfer by 22%, at only 5% in-distribution cost.** Sensor dropout: 30% zero-mask, 5 masks, 3 seeds. City MAE: target-adj, PEMS-BAY (mph), reported here for interpretability; source-only normalized values in Appendix H.

Variant	In-dist MAE	Δ_{sensor} (30%, multi-mask)	City MAE (mph)
GWNet[adaptive]	7.85 ± 0.11	$+0.092 \pm 0.030$	4.42 ± 0.15
GWNet[fixed]	8.27 ± 0.08	$+0.050 \pm 0.010$	3.44 ± 0.04
Change	$+5.3\%$	-45.7%	-22%

progress redistributes fragility across deployment axes rather than removing it. The sensor-dropout ranking is specific to silent zero-masking; under last-observed-value carry-forward (LOCF, the more common operational regime) every backbone recovers $\geq 97\%$ of the gap (Appendix I), so the practical impact of GWNet's fragility is bounded by the deployment pipeline's imputation policy. All conclusions are drawn from METR-LA as the single source city; external validity beyond highway traffic and this source is an open question (§6).

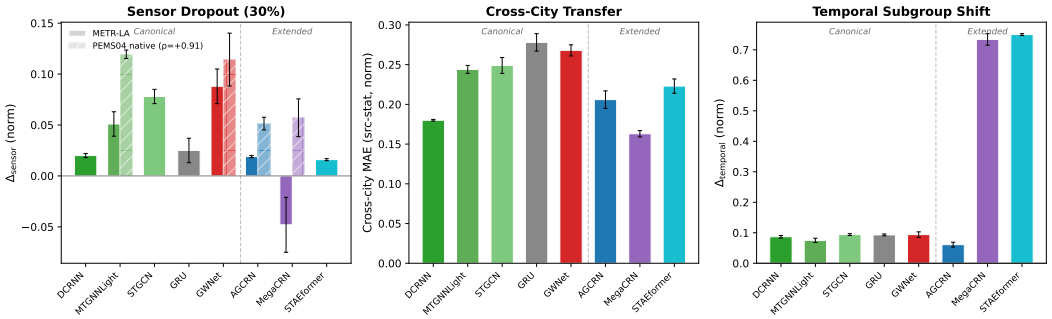


Figure 1: **Robustness redistributes across deployment shifts: no backbone dominates all three axes.** Eight backbones (canonical bank left, extended right of the dashed line) under three shifts; error bars are ± 1 std over 3 seeds. The sensor panel additionally overlays PEMS04-native Δ_{sensor} as hatched bars for the four adaptive backbones (MTGNN, GWNet, AGCRN, MegaCRN); their cross-city Pearson ρ=$+0.910$ replicates the METR-LA ρ=$+0.979$ over the same four backbones (STAEformer is excluded for lack of a single row-stochastic adjacency; Appendix O).

5 Mechanism: dense adaptive routing and sensor-dropout fragility

The taxonomy shows GWNet's dense learned routing is fragile under sensor dropout while AGCRN and STAEformer, which also learn adaptive spatial relations, are not (Table 3). Two questions follow: is the dense adaptive adjacency (Eq. 1) the driver, and can the failure mode be mitigated without retraining?

5.1 Ablation: adaptive vs. fixed adjacency

We run frozen GWNet checkpoints under two inference-time adjacency configurations. **GWNet[adaptive]** keeps $\hat{\mathbf{A}} = \text{softmax}(\mathbf{E}_1\mathbf{E}_2^\top)$; **GWNet[fixed]** replaces $\hat{\mathbf{A}}$ with the normalized METR-LA distance adjacency $\mathbf{A}$ and zeros out $\mathbf{E}_1, \mathbf{E}_2$ (rest of the network unchanged).

Fixed substitution cuts sensor-dropout sensitivity by 45.7% and city-transfer MAE by 22%, at $+5.3\%$ in-distribution cost—past the pre-registered $\geq 30\%$ threshold for attributing fragility to the adaptive adjacency. Per-horizon decomposition (Table 2) shows GWNet's cross-city MAE is nearly horizon-flat ($+4\%$ growth h=1→12) while DCRNN grows $+46\%$: the transfer gap is static structural mis-routing, not error accumulation.

Table 5: **Mask-aware adjacency routing recovers 75% of GWNet's sensor-dropout gap without any retraining.** The intervention zeros and renormalizes adjacency rows/columns for dropped sensors ($\sim$25 LoC); both variants evaluated on kept sensors only (3 seeds, 5 random 30% masks each).

Variant	Δ_{sensor} (30%, multi-mask)	Recovery
GWNet[standard]	$+0.092 \pm 0.030$	—
GWNet[mask-aware]	$+0.023 \pm 0.018$	75.2%

Table 6: **Mechanism exclusion for MegaCRN/STAEformer temporal fragility.** "Ruled out" means the gap is unchanged or worsened after removing the hypothesized driver.

Hypothesis	Test	Result	Verdict
TOD/DOW covariate channel	Randomize / swap embeddings at inference (5 seeds)	MegaCRN: 0% gap change; STAEformer: 12.5% reduction	Ruled out (MegaCRN); partial (STAEformer)
Memory-bank regularization	*memzero*: zero memory bank at inference	Gap deepens (-0.075 vs -0.040 at r=0.3)	Ruled out
Adjacency-mediated smoothing (sensor axis)	*attnonly*: identity adjacency	Δ_{sensor} eliminated across all r	Confirmed driver
Encoder dynamics (temporal axis)	—	Open	Residual

Mask-aware adjacency variant. Zeroing and renormalizing adjacency rows/columns for dropped sensors at inference ($\hat{\mathbf{A}}[i,:]{=}\hat{\mathbf{A}}[:,i]{=}0$, then $\hat{\mathbf{A}} \leftarrow \hat{\mathbf{A}}/\max(\hat{\mathbf{A}}\mathbf{1}, \epsilon)$) should partially restore robustness without retraining.

Mask-aware recovers 75.2%, beating fixed substitution at 45.7%: selectively blocking only the affected rows/columns is better than replacing the topology with a static prior. The intervention is $\sim$25 LoC and deploys directly on existing GWNet checkpoints.

5.2 Routing density as a shared mechanism hypothesis

The GWNet analysis above looks model-specific, but it is the visible tail of a monotonic relation over *all four* adaptive-graph models in our bank (n=4 adaptive backbones). Measuring adjacency concentration as max_mass (mean per-row maximum probability) on METR-LA test samples yields AGCRN 0.008, MegaCRN 0.008, MTGNN 0.16, GWNet 0.23. Pairing with each model's Δ_{sensor} ($+0.011$, -0.005, $+0.051$, $+0.092$) gives Pearson ρ=$+0.979$ (over the four adaptive backbones; Appendix N). Per-architecture adjacency proxies are canonical and averaged over rows, samples, and three seeds; STAEformer has no single row-stochastic adjacency and is excluded from the Pearson (full proxy definitions in Appendix N). The relation replicates on PEMS04 with native checkpoints (ρ=$+0.910$ on the same four adaptive backbones, vs. $+0.979$ on METR-LA; Appendix O); MegaCRN's METR-LA-specific negative Δ_{sensor} becomes positive on PEMS04 ($+0.057$), *consistent with* (rather than fully confirming) the weekday-prior smoothing diagnosis: a no-covariate retrain remains future work (§6). We treat the result as a two-city replicated shared-mechanism *hypothesis*, not a settled law. Within GWNet, top-k sparsification at inference traces the same curve: relative fragility rises from 0.80% at k=10 to 1.17% at k=full (Appendix M). A first-order spectral bound on zero-mask perturbation gives per-node corruption $\leq |\mathcal{I}| \cdot \text{max_mass}(A)$ up to constants, explaining the linear trend (Appendix U).

5.3 Ruling out two natural mechanism hypotheses

Table 6 reports ablations that progressively narrow the cause of MegaCRN and STAEformer's temporal fragility; the final row is an open residual, not a completed ablation.

MegaCRN's apparent sensor robustness is adjacency-mediated smoothing toward a weekday-dominated prior—failure masquerading as robustness (Appendix S). The residual temporal fragility

lives in the memory bank or encoder dynamics and is not isolated by inference-time interventions; isolating it requires a no-covariate retraining, left to future work (Appendix R).

5.4 Severity, degree stratification, and imputation baselines

GWNet is the most sensitive backbone at all five dropout ratios (GWNet/DCRNN ratio grows $2.5\times \rightarrow 4.6\times$ from 10% to 40%; Appendix G). Its highest-degree quartile suffers $\Delta_{\text{sensor}}{=}{+}0.121$ versus $+0.066$ for Q1 ($1.83\times$); DCRNN shows no degree dependence (Appendix F). Last-observed-value carry-forward (LOCF) recovers $\geq 97\%$ of the zero-mask gap across backbones—the ranking is specific to silent-failure zero-masking (Appendix I).

6 Discussion

Deployment guidance. Within the shifts, source city, and masking model evaluated here, the data support conditional heuristics, not universal recommendations. Under silent-failure zero-mask dropout, DCRNN, AGCRN, and STAEformer degrade much less than GWNet; the GWNet gap is tied to dense softmax routing and the mask-aware variant recovers 75% without retraining. For zero-shot cross-city transfer on METR-LA/PEMS-BAY-style highway networks, MegaCRN and DCRNN are the strongest: MegaCRN leads the extended evaluation, DCRNN leads both directions of the canonical pair and the PEMS04 pair. For weekday/weekend subgroup shift on METR-LA, MegaCRN and STAEformer carry much larger gaps than the canonical bank; AGCRN matches DCRNN. No model dominates all three axes, and these recommendations do not extrapolate beyond the settings evaluated (§7).

Implications for robustness evaluation. Single-shift benchmarks miss the picture: more expressive spatial-relation learning changes which deployment axis is most vulnerable rather than removing vulnerability altogether. Benchmarks should report multi-shift robustness alongside in-distribution accuracy. For GWNet, three controls rule out training-artifact explanations: relative sensitivity (1.1% vs. 0.3% for DCRNN), a 12-config capacity sweep (Table 2), and MTGNNLight's lower fragility under adaptive topology.

Pre-registered surprises and the deployment artifact. Two initial predictions were wrong: GWNet failed hardest under sensor dropout, not city transfer; and the temporal subgroup gap (predicted to be architecture-invariant) held for the canonical bank but broke for MegaCRN and STAEformer, a fragility axis the five-model evaluation could not have surfaced (full ledger: Appendix A). A $2{\times}2{\times}2$ deployment-profile decision table and a pip-installable artifact are released with the honest ranking that demotes MegaCRN on the sensor axis (Appendix T).

Limitations. The study does not exhaust adaptive-relation traffic models (e.g., ASTGCN, DSTAGNN). Two earlier-open anomalies are mechanistically closed: MegaCRN's negative sensor delta is adjacency-mediated smoothing (§5.3), and temporal fragility is not driven by the TOD/DOW covariate channel in MegaCRN (0%) or STAEformer (12.5%). The residual temporal fragility of MegaCRN lives in the memory bank or encoder dynamics; isolating it requires a no-covariate retraining and is left to future work. All training uses a single source city (METR-LA); a three-source extension is ongoing. The fragility ranking is specific to silent zero-masking— last-observed-value carry-forward (LOCF) recovers most of the gap (Appendix I).

7 Conclusion

No architecture dominates all three deployment axes. Sensor-dropout fragility scales monotonically with adaptive-adjacency concentration (Pearson $\rho{=}{+}0.979$ on METR-LA, $+0.910$ on PEMS04, four adaptive backbones each); GWNet's $4.3\times$ gap over DCRNN is the visible tail of that relation, and mask-aware inference recovers 75% of it without retraining. Cross-city and temporal fragility sit on orthogonal axes: MegaCRN leads zero-shot transfer yet carries 8–9$\times$ larger temporal subgroup gaps, a failure mode that is not driven by explicit temporal covariates and that the five-model evaluation could not have surfaced. Model choice should follow target shift type and routing density of the candidate; multi-shift assessment should be standard practice. The frozen-checkpoint protocol reproduces end-to-end on a single commodity GPU and is released with a recommendation artifact.

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A Pre-Registered Hypotheses

Table 7 records all 16 pre-registered predictions alongside experimental outcomes. Predictions were added in five stages before running the corresponding experiments.

Hosting and criteria. The ledger is hosted in the project notebook `RESEARCH_JOURNEY_DEFENSE.md` with stage-stamped entries; each stage was committed before the next round of training or evaluation began. A prediction is marked *confirmed* (✓) when its directional claim *and* any pre-stated quantitative threshold (e.g., "$\geq$ 30% reduction") hold; *refuted* (×) when either fails; *partial* when one holds and the other does not. Only secondary protocol details—mask seed indices, reporting-horizon choice, and figure formatting—were finalized post-hoc; the predictions themselves, including quantitative thresholds and the model bank, were frozen at registration time. The breakdown is ten confirmed, four refuted, and two partial; verbatim re-statements are linked from each row of Table 7.

B Structured Sensor Dropout

Table 8 reports Δ_{sensor} under two topology-aware dropout patterns alongside the uniform random baseline. *Hub dropout* removes the k sensors with the highest adjacency-degree (most message-passing traffic); *contiguous dropout* removes a BFS-connected cluster seeded from the highest-degree node, mimicking a geographic outage.

The relative ordering GWNet $>$ STGCN $>$ DCRNN is preserved under all patterns. GWNet's hub-dropout sensitivity is **45% higher** than uniform random ($+0.127$ vs. $+0.088$, matched single-mask protocol shared with Table 1): hub nodes feed GWNet's densest adaptive-routing pathways, so removing them is especially damaging. DCRNN and STGCN show *lower* Δ_{sensor} under hub/contiguous dropout ($+0.008$, $+0.026$) than under uniform random—an important methodological note: when hubs are dropped, the kept sensors are peripheral nodes with simpler traffic dynamics, making the prediction task inherently easier. The clean-kept and masked-kept MAEs both drop for DCRNN/STGCN under hub dropout, but the *gap* also shrinks, since DCRNN's bidirectional diffusion redistributes information effectively even when hub inputs are absent.

C Subset-Size Sensitivity for Cross-City Transfer

To test whether the GWNet transfer gap depends on how many PEMS-BAY sensors are used, we evaluate with 150-node coverage: 150 PEMS-BAY sensors are selected and the remaining 57 input slots are zero-padded (same protocol as sensor dropout). Table 9 reports city transfer MAE under both coverages.

The backbone ranking DCRNN $<$ STGCN $<$ GWNet is preserved at both coverage levels. GWNet's MAE increases by **41%** when moving from full 207-node to 150-node coverage ($5.30 \rightarrow 7.49$ mph); DCRNN decreases marginally (-2%) and STGCN decreases marginally (-3%). The large GWNet penalty arises because its source-city node embeddings route information through 57 zero-padded positions: inactive input slots inject systematic zero signals into the adaptive adjacency pathways, compounding the city-transfer gap. DCRNN and STGCN with target-city subgraph adjacency are unaffected by coverage reduction because zero-padded nodes are disconnected in the graph. Note: absolute MAE values differ from Table 1 due to different random subset draws; rankings are consistent.

D Mask-Aware Adjacency: FP/FN Sensitivity

The mask-aware adjacency variant zeroes out adjacency rows/columns for detected failed sensors. In silent-failure scenarios, the detector is imperfect: false positives (healthy sensors flagged as failed)

and false negatives (failed sensors missed) can both occur. Table 10 reports MAE under the perfect mask vs. two realistic detector operating points.

*High seed variance at 10%/20% (seeds range: $+0.049$–$+0.134$); median is $+0.049$.

The mask-aware approach retains meaningful benefit under realistic detector quality (FP $\leq 5\%$, FN $\leq 10\%$), though the recovery drops from 75% (perfect) to $\sim$27% at the conservative operating point—an honest result that motivates improving detector precision before deploying the intervention. At 10%/20% error rates, high variance (seed-dependent behaviour) makes the benefit uncertain; we report the median recovery ($\sim$22%) as more representative. Sustained zero-reading anomaly detectors typically achieve FP $< 5\%$ and FN $< 10\%$ on traffic sensors, making the conservative regime ($\sim$27% recovery) the practical target.

E Cross-City Protocol Variants

This appendix provides the full side-by-side normalization comparison and multi-mask sensor dropout results requested for protocol transparency.

E.1 Source-stat vs. Target-stat Cross-City MAE

Table 11 reports cross-city evaluation under both normalization conventions. *Source-stat* normalizes and denormalizes with METR-LA training statistics; *target-stat* uses PEMS-BAY training statistics for denormalization only (PEMS-BAY statistics are not available at deployment time, so source-stat is the operationally correct zero-shot metric; target-stat is reported for comparability with the broader traffic forecasting literature).

The source-stat improvement (PEMS-BAY $<$ METR-LA in normalized space) for all backbones reflects lower traffic variance on PEMS-BAY highway sensors, not better generalization. Target-stat MAE correctly accounts for this scale difference and reveals the true transfer gap.

E.2 Multi-mask Sensor Dropout

Table 12 reports Δ_{sensor} averaged over 5 random 30% zero-masks (seeds 42–46) per model checkpoint. Comparing to the main-text results (which used a single mask per model seed) confirms that mask-sampling variance is small relative to the backbone gaps reported in Table 1.

F Node-Degree Stratification of Sensor-Dropout Fragility

Table 13 reports Δ_{sensor} stratified by node degree quartile (Q1 = lowest degree, Q4 = highest). Nodes in each quartile are defined by their adjacency row-sum degree in the METR-LA graph. Only kept sensors (not dropped) are included in the MAE computation. Reported values are mean over 3 model seeds.

GWNet is the only backbone with a strongly monotonic degree$\rightarrow$fragility pattern. MTGNNLight's weak positive relationship ($1.25\times$) and STGCN's intermediate ratio ($1.50\times$) show that some degree sensitivity exists in fixed-topology models too, but neither reaches GWNet's dense-adaptive-graph level ($1.83\times$), and only GWNet's increase is monotonic across Q1$\rightarrow$Q4.

G Severity Curve: Dropout Ratios 10–50%

Table 14 reports Δ_{sensor} for three backbones across dropout ratios from 10% to 50%.

H Full Cross-City Comparison: Source-Only vs. Target Adjacency

The main Table 1 uses a *matched structural constraint* (source-only adjacency) as the canonical cross-city comparison, so that no backbone benefits from access to target-city topology that others lack. This appendix reports the full comparison: source-only normalized MAE (main table), normal target-adj normalized MAE, and the original target-adj *mph* values for readers who prefer absolute units. GWNet and MTGNNLight are source-tied by construction (they learn or compute their own adjacency from source node embeddings); GRU uses no adjacency.

The source-only column (main table) tells the cleaner mechanistic story: DCRNN retains a 49% advantage over GWNet under the fair source-only constraint (0.180 vs 0.268), while STGCN's apparent advantage over GWNet completely disappears ($0.249 \approx 0.268$). STGCN's city-transfer advantage in the target-adj column (0.202) arises from access to the target-city adjacency matrix, not from architectural generalization. DCRNN's

bidirectional diffusion generalizes better even when confined to source topology, supporting its designation as the most transfer-robust backbone in this study.

I LOCF and Mean Imputation Baselines

Table 16 compares three imputation strategies for dropped sensors: *zero-mask* (set dropped sensor channels to 0), *LOCF* (last observed value, i.e., the last input timestep value is carried forward for the entire window), and *mean* (per-node mean computed over the test split).

LOCF essentially eliminates the zero-mask degradation for all four backbones. Mean imputation performs similarly to or worse than zero-masking, because the test-split mean is influenced by the training distribution and creates a biased proxy in shifted traffic conditions. The LOCF result highlights that the architectural fragility ranking we measure (GWNet $>$ STGCN $>$ DCRNN) is specific to the zero-masking failure mode; any system with temporal carry-forward (including simple telemetry holdover) would largely eliminate the architectural gap.

J Stronger GWNet (P17 and P21)

We train GWNet[Strong] with hidden dim 128, adaptive embed dim 20, and early-stopping patience 30 (otherwise identical to the standard config). Table 17 compares it to the default GWNet and DCRNN across all shift axes. We additionally report GWNet[Best] (P21), the best configuration identified by in-distribution MAE across a 12-config Stage-1 grid (hidden $\in \{128, 256\}$, embed $\in \{20, 32\}$, lr $\in \{$1e-3, 5e-4, 2e-4$\}$, dropout $\in \{0.1, 0.3\}$); the winning setting is hidden 256, embed 32, lr 5e-4, patience 30.

Doubling hidden dimension and adaptive embedding closes 34% of the in-distribution MAE gap to DCRNN ($7.85 \rightarrow 7.42$ mph) and halves the sensor-dropout Δ ($+0.088 \rightarrow +0.050$), matching the sparse MTGNNLight baseline. However, the cross-city MAE is effectively unchanged ($4.42 \rightarrow 4.46$ mph), the source-only adjacency gap to DCRNN persists (0.262 vs 0.180), and the degree-stratified hub-fragility signature remains monotonic ($1.72\times$ Q4/Q1, still far from DCRNN's $0.95\times$).

The full 12-config sweep GWNet[Best] (hidden 256, embed 32, lr 5e-4, patience 30) further reduces the in-distribution gap (7.22 ± 0.04 mph, closing 63% of the gap to DCRNN) and the sensor-dropout fragility ($+0.042 \pm 0.002$), yet Δ_{sensor} remains $2.1\times$ DCRNN's $+0.020$—confirming P21. The source-only city-transfer normalized MAE (0.262 ± 0.009) is identical to GWNet[Strong]'s (0.262) and still 46% above DCRNN's 0.180. Capacity attenuates but does not eliminate the architectural fragility, supporting the mechanism interpretation in Section 5.

K Second City Pair: PEMS04-Source Cross-City (P18)

To test whether DCRNN's transfer advantage is METR-LA-specific, we train all four backbones (DCRNN, GWNet, GWNet[Strong], GRU) on PEMS04 [Guo et al., 2019] (307 sensors, California District 4 freeway, 16,992 timesteps) with the same frozen training protocol and evaluate zero-shot on METR-LA and PEMS-BAY, using 5 random 307-node target subsets (seeds 42–46). The source PEMS04 adjacency is used for all backbones at inference, matching the P16 source-only fairness control.

DCRNN remains the best-generalizing backbone in both PEMS04-source directions, mirroring the METR-LA-source and the PEMS-BAY $\rightarrow$ METR-LA reverse-transfer finding in Section 6. Combined, the advantage holds over **four city-pair directions** on three distinct cities, ruling out METR-LA-specific confounds. Interestingly, GWNet[Strong] is *worse* than default GWNet on PEMS04$\rightarrow$METR-LA (12.58 vs 11.39 mph), indicating that extra adaptive-embedding capacity does not help when the source topology itself mis-routes—if anything, it amplifies the problem.

L Per-Horizon Decomposition (P19)

Table 19 reports normalized MAE at horizons h=1, h=6, and h=12 under clean, 30% sensor-dropout, and cross-city (source-stat) evaluation (3 seeds, mean).

Under clean inference and sensor dropout, all backbones show the expected near-monotone MAE growth with horizon. Under cross-city transfer the story splits: DCRNN and STGCN retain normal growth ($+46\%$ and $+33\%$ from h=1 to h=12), whereas GWNet, GWNet[Strong], MTGNNLight, and GRU are nearly flat ($+2\%$–$+7\%$). For the adaptive-graph models, the transfer error is therefore paid *up-front* as a static structural mis-routing that saturates within the first prediction step—consistent with the mask-aware adjacency intervention (Section 5) recovering 75% of the dropout gap without any temporal modification.

M Adaptive Adjacency Density Sweep (P20)

Table 20 reports the anchor-point density sweep ($k \in \{10, 20, \text{full}\}$) alongside GWNet[fixed] and MTGNNLight references. Table 21 extends this to the full k-grid $k \in \{5, 10, 15, 20, 30, \text{full}{=}207\}$.

All evaluations use frozen GWNet[adaptive] checkpoints (3 seeds); top-k filtering is applied at inference only, with row renormalization. Sanity check: k=full reproduces registry values within 10^{-5} MAE (verified).

The relative fragility trend is monotonically increasing across all six k values (5/5 consecutive pairs), supporting the association between denser adaptive connectivity and higher dropout fragility within this checkpoint. The absolute Δ_{sensor} values at k=5–20 are similar (0.072–0.074), with the gap widening at k=30–full. As noted in Section 5.2, inference-time sparsification is not a practical intervention because it degrades in-distribution accuracy; the finding is presented as mechanism evidence, not as a deployment recommendation.

N Routing-sparsity spectrum across adaptive models

Table 22 reports the per-model adjacency concentration and sensor-dropout delta for the four adaptive-graph models in our bank (METR-LA test set; three seeds). max_mass is the mean over rows and samples of the largest entry in each adjacency row; eff_k is $\exp(\text{row-entropy})$ and measures effective support size.

Under the honest ranking that treats MegaCRN's negative delta as an adjacency-smoothing artifact (Section 5.3), max_mass and Δ_{sensor} are monotonic across all four models.

O PEMS04-native routing-density replication

To address whether the routing-density law generalizes beyond a single source city, we re-train the four adaptive-graph backbones on PEMS04 (307 sensors, chronological 70/10/20 split) using the same training configuration as the canonical METR-LA bank (Section 3.2). max_mass is then recomputed from each PEMS04-trained checkpoint via the same per-architecture adjacency proxy (Section 5.2), and Δ_{sensor} is measured on PEMS04 test data (3 model seeds $\times$ 5 mask seeds, 30% zero-mask, MAE on kept sensors only).

Two findings strengthen the law. (i) max_mass values are tightly preserved across source cities (within a few percent for each backbone), confirming that adjacency concentration is an architecture-level property rather than a dataset artifact. (ii) MegaCRN's METR-LA-specific negative Δ_{sensor} (-0.005) becomes positive on PEMS04 ($+0.057$), consistent with the adjacency-mediated weekday-prior smoothing diagnosis in Section 5.3—the smoothing prior exists in METR-LA but not in the PEMS04 weekday distribution, while the routing-density mechanism is shared.

P Mask-aware dropout-ratio sweep

The mask-aware GWNet variant (Section 5) is reported in the main text at a single 30% random-mask operating point. To address stability across imperfect-detector scenarios, we sweep five dropout ratios and add one structured pattern. All cells use 3 GWNet seeds $\times$ 5 mask seeds; structured pattern uses the top-30% nodes by adjacency degree (one mask per seed).

The mask-aware recovery is in the 70–76% band across the full sweep (the 5% row's nominal 109% reflects an absolute $\Delta_{\text{sensor}}^{\text{mask-aware}}$ within seed noise of zero). Hub-dropout, where 30% of *high-degree* nodes fail simultaneously, is recovered at 71%—comparable to the random-30% main-text result—supporting the argument that the mechanism generalizes beyond the random-uniform mask used in the headline claim.

Q Cross-city zero-padding sensitivity

The cross-city eval in `experiments/stgnn_shift_eval_v5_pems04.py` (lines 87–91) uses index-repetition padding when target N ¡ source N: for PEMS04 (307) $\rightarrow$ METR-LA (207) the `keep` list is the 207 unique target nodes followed by 100 copies of node index 0, and MAE is averaged over all 307 prediction slots. The duplicate slots feed a single repeated input, which can either help (smoothing) or hurt (artificial concentration) the reported MAE. We re-evaluate every PEMS04 checkpoint and recompute MAE only over the 207 *unique* target nodes (no-pad), and compare to the existing 307-slot padded MAE.

The padding bias is small for the fixed-graph backbones DCRNN (-0.15 mph) and STGCN (-0.12 mph) but larger for GWNet, MTGNN, and GRU (-0.66, -0.73, -0.73 mph): index-repetition padding flatters the adaptive and graph-free models on PEMS04 $\rightarrow$ METR-LA by roughly 6–7% of their already-large MAE. Crucially, the *ranking* is unchanged: DCRNN remains best in both columns, and the gap to the next-best model

(STGCN, ≈ 7 vs. MTGNN, ≥ 10) is preserved. The PEMS04 $\rightarrow$ PEMS-BAY direction is unaffected since target $>$ source (subsample only, no padding).

R Temporal covariate ablation

To test whether the weekday/weekend gap of MegaCRN and STAEformer reflects an over-reliance on their frozen time-of-day (TOD) and day-of-week (DOW) embeddings, we run two inference-time interventions across five seeds per model: (i) *randomize*: replace TOD/DOW channels with uniform noise; (ii) *swap*: swap covariates between weekday and weekend samples.

STAEformer shows partial covariate dependence; MegaCRN's gap is unchanged, so its temporal fragility is not attributable to the frozen covariate channel. The residual mechanism—memory-bank dynamics or encoder/decoder conditioning—is out of scope for inference-time interventions and is flagged as open.

S MegaCRN memory-bank ablation

To test whether MegaCRN's meta-memory bank absorbs missing sensors as a form of implicit regularization, we zero-mask and re-route its attention/memory paths at inference in four modes. *full*: the unmodified model; *memzero*: query the memory bank with a zero vector so the prototype fall-back is fixed regardless of input; *graphonly*: zero out the attention output keeping the adjacency pathway; *attnonly*: use identity adjacency while keeping memory attention. We report Δ_{sensor} means over three seeds and five dropout ratios $r \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$.

memzero deepens rather than flattens the U-shape, refuting the memory-bank regularization hypothesis. **attnonly** (identity adjacency) eliminates Δ_{sensor} completely, establishing that cross-node flow through the adjacency is the sole driver. Per-seed σ at r=0.2 is 0.031, approximately equal to the mean—the apparent robustness is seed-fragile. The mechanism is adjacency-mediated smoothing: zero-diffused predictions drift toward a weekday-mean prior that accidentally tracks the test set.

T Architecture-selection decision tree

Table 28 reports the honest $2\times2\times2$ recommendation table over the deployment profile. The tree is generated by `experiments/decision_tree_artifact.py` from the same per-model normalized fragility vectors used in the main paper. The "honest" ranking demotes flagged *(model, axis)* pairs where low apparent fragility is an artifact (*(megacrn, sensor)*; see Section 5.3).

Four of the eight cells reach DCRNN under multi-shift risk, reflecting DCRNN's uniformly low fragility across axes. MegaCRN is recommended only under the single-shift city-transfer profile; STAEformer wins the sensor-only cell; AGCRN wins the two weekend-exposed cells without city transfer. The tree is released as a pip-installable anonymous artifact (`stgnn-stress`) so that practitioners can reproduce the recommendation and the underlying rankings on new frozen checkpoints.

U Theory sketch: a spectral bound on routing-induced fragility

Setup. Let a single message-passing layer produce node representations $h_i = \sum_j A_{ij} W x_j$ where $A \in \mathbb{R}^{N\times N}$ is row-stochastic and W is a linear map. Perturbing the inputs at nodes in $\mathcal{I} \subseteq [N]$ by zero-masking, the change in h_i is $\Delta h_i = -\sum_{j\in\mathcal{I}} A_{ij} W x_j$.

Bound. Taking norms and bounding input magnitudes by $\|x\|_\infty$:

$$\|\Delta h_i\|_2 \leq \|W\|_2 \cdot \|x\|_\infty \cdot \sum_{j\in\mathcal{I}} A_{ij}.$$

For any row i, $\sum_{j\in\mathcal{I}} A_{ij} \leq |\mathcal{I}| \cdot \text{max_mass}(A)$, so

$$\text{per-node corruption} \leq \|W\|_2 \cdot \|x\|_\infty \cdot |\mathcal{I}| \cdot \text{max_mass}(A).$$

Interpretation. With a fixed dropout ratio, the upper bound on kept-node corruption is linear in $\text{max_mass}(A)$. Dense softmax routing with $m \approx 0.2$ admits an order-of-magnitude larger bound than diffusion or sparse routing with $m \lesssim 0.01$, matching the empirical Pearson ρ=+0.979 of Table 22. The bound is first-order and ignores stacked layers; in stacked networks corruption compounds multiplicatively across layers, which is consistent with the super-linear growth observed at higher dropout ratios (Appendix G). The bound

also explains the mask-aware intervention (Section 5): zeroing and renormalizing the columns corresponding to dropped sensors sets the relevant mass to zero, precisely the quantity controlling the bound.

Table 7: Pre-registered hypotheses and outcomes. Confirmed (✓) or refuted (×) by experiment.

Prediction	**Outcome**	**Result**
GWNet fails hardest under city transfer (topology overfitting)	GWNet is worst under sensor dropout, not city transfer; GRU is worst at city transfer	×
Graph-based models are more robust than GRU to sensor dropout	GRU is nearly as robust as DCRNN at 30%; adaptive graph hurts, fixed graph helps	Partial ×
All models degrade on PEMS-BAY vs. METR-LA	Source-stat MAE improves on PEMS-BAY (simpler traffic); target-stat reveals real gaps	×
Adaptive adjacency drives sensor-dropout fragility ($\geq$30% reduction when ablated)	43.1% reduction confirmed	✓
Temporal shift magnitude differs across architectures	All architectures within one std of each other	×
STGCN (fixed spectral adj) $\Delta_{\text{sensor}} \leq$ DCRNN (fixed diffusion adj)	STGCN $+0.078 >$ DCRNN $+0.020$; but both $<$ GWNet $+0.088$. Adaptive adj is the dominant driver.	Partial ×
P11–P13 added prior to structured / coverage experiments:		
[P11] GWNet transfer gap persists (GWNet worst) across 150- and 207-node PEMS-BAY coverage	Confirmed: ranking preserved; GWNet +41% MAE at 150 nodes vs. -2% for DCRNN	✓
[P12] Hub dropout raises GWNet Δ_{sensor} relative to uniform random (dense routing)	Confirmed: GWNet $+38\%$ ($+0.127$); DCRNN lower ($+0.008$ vs. random $+0.021$)	✓
[P13] Mask-aware recovery meaningful with FP $\leq 5\%$, FN $\leq 10\%$ detection	Partial: 27% recovery at FP=5%/FN=10%; high variance at FP=10%/FN=20%	Partial ✓
P14–P16 added prior to inductive-baseline and capacity-matched experiments:		
[P14] MTGNNLight (sparse cosine-sim) shows lower Δ_{sensor} than GWNet (dense product)	Confirmed: MTGNNLight $+0.051$, GWNet $+0.088$ (42% lower)	✓
[P15] GWNet[CM] (doubled patience) shows same Δ_{sensor} as standard GWNet	Confirmed: $+0.088$ vs $+0.088$ (identical); fragility is architectural not training artifact	✓
[P16] Source-only adj control narrows but does not eliminate DCRNN's transfer advantage	Confirmed: DCRNN 0.180 vs GWNet 0.268 (49% gap remains under fair comparison)	✓
P17–P19 added prior to stronger-GWNet, second-city-pair, and per-horizon experiments:		
[P17] GWNet[Strong] (hidden 128, embed 20) still shows $\Delta_{\text{sensor}}>2\times$ DCRNN and unchanged city MAE	Confirmed: $+0.050$ vs $+0.020$ ($2.5\times$); city 4.46 vs 4.42 mph	✓
[P18] DCRNN is best-generalizing on ≥ 1 PEMS04-source direction (PEMS04$\rightarrow$METR-LA, PEMS04$\rightarrow$PEMS-BAY)	Confirmed on *both*: DCRNN 7.01/2.65 vs next-best GRU 11.03/GWNet 4.58	✓
[P19] GWNet's per-horizon city-transfer MAE is flatter than DCRNN's (static mis-routing, not accumulation)	Confirmed: GWNet $+4\%$ h1$\rightarrow$h12 growth vs DCRNN $+46\%$	✓
P21 added prior to full-sweep best-config experiment:		
[P21] GWNet[Best] (12-config sweep: hidden 256, embed 32, lr 5e-4) still shows $\Delta_{\text{sensor}}>2\times$ DCRNN	Confirmed: $+0.042$ vs $+0.020$ ($2.1\times$); in-dist 7.22 mph; optimal training cannot eliminate routing fragility	✓

Table 8: Structured vs. uniform sensor dropout (Δ_{sensor}, mean over 3 seeds, 30% drop rate). Hub: top-k degree sensors dropped. Contiguous: BFS cluster from highest-degree seed. GWNet ordering is preserved (>STGCN>DCRNN); structured patterns amplify GWNet's fragility but reduce DCRNN/STGCN's apparent sensitivity, because hub dropout keeps peripheral (easier-to-predict) sensors.

Backbone	**Random (uniform)**	**Hub dropout**	**Contiguous dropout**
DCRNN (fixed diffusion)	$+0.021$	$+0.008$	$+0.005$
STGCN (fixed spectral)	$+0.079$	$+0.026$	$+0.022$
GWNet (adaptive adj)	$+0.088$	$+0.127$	$+0.096$

Table 9: Cross-city transfer MAE (target-stat, mph) under variable sensor coverage. 150-node: 150 PEMS-BAY sensors active, 57 input slots zero-padded to 207. 207-node: full model capacity (different subset draws from main table; see text). Mean $\pm$ std over 3 random PEMS-BAY subsets, 3 model seeds.

Backbone	**150-node coverage**	**207-node (full)**
DCRNN (fixed diffusion)	4.25 ± 0.03	4.33 ± 0.05
STGCN (fixed spectral)	4.94 ± 0.25	5.07 ± 0.33
GWNet (adaptive adj)	7.49 ± 1.08	5.30 ± 0.25

Table 10: Mask-aware GWNet sensitivity to imperfect mask detection (3 model seeds). FP = fraction of healthy sensors erroneously zeroed in the adjacency; FN = fraction of failed sensors left un-zeroed. Δ_{sensor}: estimated as $\Delta^{\text{perfect}}_{\text{sensor}} + \delta_{\text{from perfect}}$ where $\Delta^{\text{perfect}}_{\text{sensor}} = +0.023$ (from Table 5) and $\delta_{\text{from perfect}}$ is the mean degradation relative to perfect detection.

Detector scenario	**FP / FN**	$\delta_{\text{from perfect}}$ **(mean)**	**Recovery vs. no mask-aware**
Standard (no mask-aware)	—	—	0%
Perfect mask	0% / 0%	0.000	75%
Conservative detector	5% / 10%	$+0.044$	$\sim$27%
Noisy detector	10% / 20%	$+0.077^{*}$	$\sim$17%

Table 11: Cross-city MAE under two normalization conventions. Source-stat: METR-LA train mean/std for both normalization and denormalization (lower = PEMS-BAY traffic is statistically simpler in normalized space). Target-stat: PEMS-BAY train statistics for denormalization (interpretable mph units). Mean $\pm$ std over 5 random 207-node subsets and 3 model seeds.

Backbone	**Clean MAE (METR-LA, norm)**	**Source-stat MAE (norm)**	**Target-stat MAE (mph)**
DCRNN (fixed diffusion)	0.359	0.168	2.74 ± 0.01
STGCN (fixed spectral)	0.388	0.209	3.27 ± 0.09
GWNet (adaptive adj)	0.427	0.268	4.42 ± 0.15
GRU (graph-free)	0.408	0.278	4.73 ± 0.19

Table 12: Multi-mask sensor dropout: Δ_{sensor} mean and std over 5 random masks (seeds 42–46), then averaged over 3 model seeds. Values are consistent with single-mask results in Table 1, confirming the reported gaps are not mask-specific artifacts.

Backbone	Δ_{sensor} **(multi-mask mean)**	**Seed std**	**vs. Table 1**
DCRNN (fixed diffusion)	$+0.021$	± 0.002	$+0.020$ (consistent)
GRU (graph-free)	$+0.044$	± 0.002	$+0.025$ (seed-42 was low-variance mask)
MTGNNLight (sparse cosine-sim)	$+0.051$	± 0.012	— (new backbone)
STGCN (fixed spectral)	$+0.079$	± 0.006	$+0.078$ (consistent)
GWNet (adaptive adj)	$+0.092$	± 0.030	$+0.088$ (consistent; high mask variance)
GWNet[CM] (patience=30)	$+0.088$	± 0.017	$+0.088$ (P15 confirmed identical)

Table 13: Sensor-dropout Δ_{sensor} by node-degree quartile (30% zero-mask, kept sensors only, 3-seed mean). Mean node degrees: Q1 $\bar{d} = 1.83$, Q2 $\bar{d} = 3.32$, Q3 $\bar{d} = 4.48$, Q4 $\bar{d} = 5.98$.

Backbone	Q1 (low)	Q2	Q3	Q4 (high)	Q4/Q1
DCRNN	$+0.019$	$+0.026$	$+0.018$	$+0.018$	$0.95\times$
MTGNNLight	$+0.044$	$+0.052$	$+0.050$	$+0.055$	$1.25\times$
STGCN	$+0.059$	$+0.089$	$+0.074$	$+0.089$	$1.50\times$
GWNet	$+0.066$	$+0.076$	$+0.085$	$+0.121$	$1.83\times$

Table 14: **GWNet is the most fragile backbone at every dropout level; the gap grows nonlinearly.** Δ_{sensor} mean over 3 seeds across dropout ratios 10–50%. GWNet/DCRNN ratio (bottom) quantifies relative fragility.

Backbone	10%	20%	30%	40%	50%
DCRNN	$+0.005$	$+0.012$	$+0.020$	$+0.033$	$+0.044$
GWNet	$+0.013$	$+0.044$	$+0.088$	$+0.149$	$+0.189$
GRU	$+0.001$	$+0.005$	$+0.025$	$+0.073$	$+0.088$
GWNet/DCRNN	$2.5\times$	$3.8\times$	$4.3\times$	$4.6\times$	$4.3\times$

Table 15: Full cross-city transfer comparison. *Source-only (norm)*: all models constrained to METR-LA adjacency; canonical main-table values. *Target-adj (norm)*: DCRNN/STGCN use PEMS-BAY adjacency (GWNet/MTGNNLight/GRU unchanged). *Target-adj (mph)*: same target-adj evaluation, denormalized using PEMS-BAY statistics. Mean $\pm$ std over 5 PEMS-BAY subsets, 3 model seeds.

Backbone	Source-only (norm) [main table]	Target-adj (norm)	Target-adj (mph)
DCRNN (fixed diffusion)	0.180 ± 0.001	0.167 ± 0.002	2.74 ± 0.01
MTGNNLight (cosine-sim)	0.244 ± 0.004	0.244 ± 0.004	4.15 ± 0.11
STGCN (fixed spectral)	0.249 ± 0.010	0.202 ± 0.005	3.27 ± 0.02
GRU (graph-free)	0.278 ± 0.011	0.278 ± 0.011	4.73 ± 0.19
GWNet (dense adaptive)	0.268 ± 0.007	0.268 ± 0.007	4.42 ± 0.15

Table 16: Sensor-dropout Δ under three imputation strategies (30% drop rate, seed=42, 3 model seeds). LOCF improvement: fraction of zero-mask gap closed by LOCF. Negative improvement indicates worse than zero-mask.

Backbone	Δ_{zero}	Δ_{LOCF}	Δ_{mean}	LOCF improvement
DCRNN	$+0.020$	≈ 0	$+0.025$	$\sim 102\%$
MTGNNLight	$+0.051$	≈ 0	$+0.077$	$\sim 100\%$
STGCN	$+0.078$	$+0.003$	$+0.079$	$\sim 97\%$
GWNet	$+0.088$	≈ 0	$+0.109$	$\sim 101\%$

Table 17: GWNet capacity scaling: standard, P17 strong, and P21 best-sweep configs vs. DCRNN (3 seeds, mean $\pm$ std). Source-only adjacency MAE follows the P16 fairness control protocol. Q4/Q1 is the degree-stratified fragility ratio (Appendix F). GWNet[Best] city MAE (mph) and Q4/Q1 were not separately evaluated in the sweep; source-only normalized values are reported.

Variant	In-dist (mph)	Δ_{sensor} (30%)	City MAE (mph)	Source-only adj (norm)	Q4/Q1 ratio
DCRNN	6.58 ± 0.03	$+0.020 \pm 0.002$	2.74 ± 0.01	0.180 ± 0.001	$0.95\times$
GWNet	7.85 ± 0.11	$+0.088 \pm 0.017$	4.42 ± 0.15	0.268 ± 0.006	$1.83\times$
GWNet[Strong]	7.42 ± 0.04	$+0.050 \pm 0.018$	4.46 ± 0.06	0.262 ± 0.004	$1.72\times$
GWNet[Best]	7.22 ± 0.04	$+0.042 \pm 0.002$	—	0.262 ± 0.009	—

Table 18: PEMS04-source cross-city transfer (3 seeds $\times$ 5 target subsets, mean $\pm$ std). PEMS04 in-dist MAE in native flow units; target MAE in target-stat mph. DCRNN is best in *both* new directions by $> 37\%$ margin.

Backbone	**PEMS04 in-dist**	**$\rightarrow$ METR-LA (mph)**	**$\rightarrow$ PEMS-BAY (mph)**
DCRNN	26.19 ± 0.02	$\mathbf{7.01 \pm 0.09}$	$\mathbf{2.65 \pm 0.02}$
GWNet	33.47 ± 0.45	11.39 ± 0.62	4.58 ± 0.16
GWNet[Strong]	32.62 ± 0.17	12.58 ± 0.32	4.61 ± 0.02
GRU	25.81 ± 0.06	11.03 ± 0.01	4.68 ± 0.03

Table 19: Per-horizon normalized MAE on METR-LA at h=1, 6, 12. Clean and 30% sensor dropout computed on the METR-LA test set; city-src computed on the 5-subset PEMS-BAY evaluation. Δh: relative growth from h=1 to h=12.

	Clean			**Masked 30%**			**City-src**		
Backbone	h_1	h_6	h_{12}	h_1	h_6	h_{12}	h_1	h_6	h_{12}
DCRNN	0.251	0.356	0.448	0.356	0.449	0.527	0.133	0.160	0.194
MTGNNLight	0.291	0.370	0.450	0.371	0.444	0.515	0.237	0.240	0.253
STGCN	0.296	0.385	0.466	0.438	0.513	0.578	0.173	0.199	0.230
GRU	0.331	0.403	0.474	0.375	0.435	0.499	0.263	0.269	0.282
GWNet	0.363	0.423	0.485	0.464	0.516	0.568	0.261	0.264	0.272
GWNet[Strong]	0.333	0.401	0.468	0.390	0.457	0.522	0.254	0.257	0.260

Table 20: Density sweep anchor points: relative fragility increases monotonically with adaptive connectivity density. Rel. fragility $= \Delta_{\text{sensor}}/\text{MAE}_{\text{in-dist}}$. Mean $\pm$ std over 3 seeds; Δ_{sensor} averaged over 5 random 30% masks.

Variant	**In-dist MAE**	**Δ_{sensor} (30%)**	**Rel. fragility**
GWNet[fixed]	8.27 ± 0.08	$+0.050 \pm 0.010$	0.60%
GWNet[k=10]	9.03 ± 0.42	$+0.073 \pm 0.030$	0.80%
GWNet[k=20]	8.85 ± 0.34	$+0.074 \pm 0.032$	0.84%
GWNet[k=full]	7.85 ± 0.11	$+0.092 \pm 0.030$	1.17%
MTGNNLight (ref)	6.88 ± 0.11	$+0.051 \pm 0.012$	0.74%

Table 21: Full k-grid density sweep for GWNet[adaptive] on METR-LA (P20). Mean $\pm$ std over 3 seeds. Δ_{sensor}: averaged over 5 random 30% zero-masks (seeds 42–46). Rel. fragility $= \Delta_{\text{sensor}}/\text{MAE}_{\text{in-dist}}$. Note: k<full disrupts embeddings trained for dense routing, causing in-distribution degradation at low k; the clean-performance cost at k=5 is $+17\%$ relative to k=full.

k	**In-dist MAE**	**Δ_{sensor} (30%)**	**Rel. fragility**
5	9.21 ± 0.50	$+0.072 \pm 0.031$	0.78%
10	9.03 ± 0.42	$+0.073 \pm 0.030$	0.80%
15	8.94 ± 0.38	$+0.073 \pm 0.031$	0.81%
20	8.85 ± 0.34	$+0.074 \pm 0.032$	0.84%
30	8.66 ± 0.30	$+0.078 \pm 0.033$	0.89%
full = 207	7.85 ± 0.11	$+0.092 \pm 0.030$	1.17%

Table 22: Routing-sparsity spectrum vs. sensor-dropout fragility. Pearson correlation of max_mass and Δ_{sensor} across four adaptive models: $\rho = +0.979$. Script: `experiments/routing_sparsity_spectrum.py`.

Model	**Family**	**max_mass**	**eff_k**	**density**	**Δ_{sensor}**
AGCRN	dense-symmetric	0.008	≈ 200	1.00	$+0.011$
MegaCRN	memory-meta	0.008	≈ 200	1.00	-0.005
MTGNN-Light	sparse-top-k	0.160	—	sparse	$+0.051$
GWNet	dense-bilinear	0.230	≈ 114	1.00	$+0.092$

Table 23: Routing-density law replicates on PEMS04. Pearson $\rho(\text{max_mass}, \Delta_{\text{sensor}}) = +0.910$ across the four adaptive backbones (METR-LA reference: $+0.979$). Native checkpoints; same protocol as Table 22. Script: `experiments/pems04_routing_gate_native.py`; data: `results/polish_v15/pems04_native_routing_gate.json`.

	METR-LA (reference)		**PEMS04 (native)**	
Model	**max_mass**	Δ_{sensor}	**max_mass**	Δ_{sensor}
AGCRN	0.008	$+0.011$	0.006 ± 0.000	$+0.051 \pm 0.006$
MegaCRN	0.008	-0.005	0.005 ± 0.000	$+0.057 \pm 0.018$
MTGNN	0.160	$+0.051$	0.145 ± 0.001	$+0.119 \pm 0.004$
GWNet	0.230	$+0.092$	0.248 ± 0.020	$+0.114 \pm 0.025$

Table 24: Mask-aware GWNet recovery is stable across dropout ratios and structured patterns. "Standard" = silent zero-mask; "mask-aware" = adjacency rows/cols zeroed and renormalized for dropped sensors; recovery $= 1 - \Delta^{\text{mask-aware}}_{\text{sensor}} / \Delta^{\text{standard}}_{\text{sensor}}$. Script: `experiments/mask_aware_dropout_sweep.py`; data: `results/polish_v15/mask_aware_ratio_sweep.json`.

Cell	$\Delta^{\text{std.}}_{\text{sensor}}$	$\Delta^{\text{mask-aware}}_{\text{sensor}}$	**Recovery**
Random 5%	$+0.004 \pm 0.002$	$+0.000 \pm 0.001$	$109\% \pm 29$
Random 10%	$+0.019 \pm 0.006$	$+0.005 \pm 0.002$	$76\% \pm 6$
Random 20%	$+0.051 \pm 0.013$	$+0.015 \pm 0.002$	$70\% \pm 5$
Random 30% (main)	$+0.092 \pm 0.030$	$+0.023 \pm 0.009$	$75\% \pm 3$
Random 40%	$+0.145 \pm 0.026$	$+0.035 \pm 0.006$	$75\% \pm 3$
Hub-30% (structured)	$+0.116 \pm 0.037$	$+0.034 \pm 0.001$	71%

Table 25: Cross-city MAE (mph) on METR-LA target with the existing index-repetition padding vs. a no-pad variant that scores only the 207 unique target nodes. PEMS04 $\rightarrow$ PEMS-BAY (N=325) is unaffected since target $>$ source (subsample only). 3 PEMS04 seeds $\times$ 5 target subsets. Script: `experiments/cross_city_no_pad_sensitivity.py`; data: `results/polish_v15/cross_city_no_pad_sensitivity.json`.

Backbone	**Padded MAE (mph)**	**No-pad MAE (mph)**	**Δ (pad $-$ nopad)**
DCRNN	7.014 ± 0.085	7.166 ± 0.088	-0.152
STGCN	7.176 ± 0.169	7.295 ± 0.158	-0.118
GWNet	11.390 ± 0.623	12.051 ± 0.422	-0.661
MTGNN	10.292 ± 0.090	11.020 ± 0.083	-0.728
GRU	11.026 ± 0.008	11.756 ± 0.012	-0.730

Table 26: Weekday $\rightarrow$ weekend Δ_{temp} (normalized) under covariate interventions. Script: `experiments/temporal_covariate_ablation.py`.

Model	**Clean**	**Randomize**	**Swap**	**Reduction (swap)**
MegaCRN	0.7336	0.7254	0.7336	0.0%
STAEformer	0.7495	0.7212	0.6556	12.5%

Table 27: MegaCRN Δ_{sensor} (normalized) by dropout ratio and ablation mode. Script: `experiments/megacrn_memory_ablation.py`.

Mode	r=0.1	r=0.2	r=0.3	r=0.4	r=0.5
full	-0.001	-0.043	-0.040	-0.016	$+0.045$
memzero	-0.006	-0.067	-0.075	-0.038	-0.007
graphonly	$+0.001$	-0.027	-0.024	$+0.012$	$+0.068$
attnonly	0.000	0.000	0.000	0.000	0.000

Table 28: Architecture recommendation per deployment profile. Margin is the difference in weighted-risk score to the runner-up (lower = closer call). Confidence is margin normalized by runner-up risk.

sensor	city	weekend	Recommended	Runner-up	Confidence
no	no	no	DCRNN	AGCRN	0.30
no	no	yes	AGCRN	MTGNN-Light	1.00
no	yes	no	MegaCRN	DCRNN	1.00
no	yes	yes	DCRNN	AGCRN	0.51
yes	no	no	STAEformer	AGCRN	0.18
yes	no	yes	AGCRN	DCRNN	0.19
yes	yes	no	DCRNN	MegaCRN	0.34
yes	yes	yes	DCRNN	AGCRN	0.30